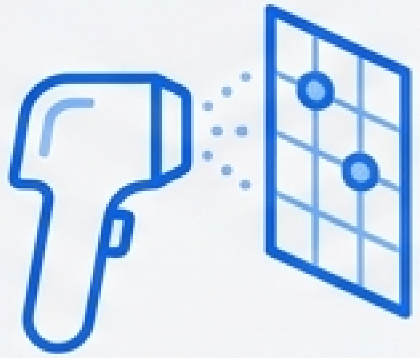
Beyond the Search Bar: Empowering Patients with AI-Assisted Dermatology



Access to health information outpaces clinical comprehension

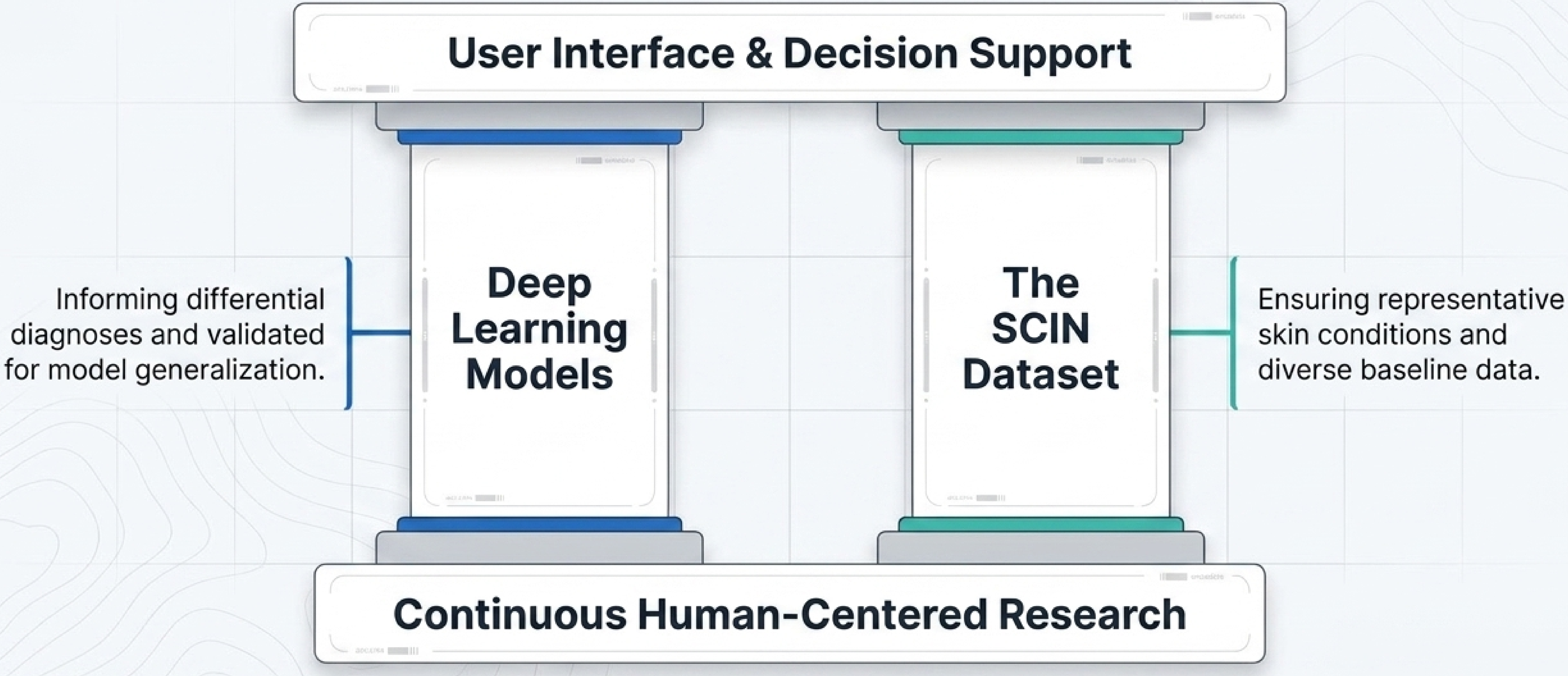
THE ACCESS

OVER 50% OF ADULTS USE THE INTERNET FOR HEALTH INFORMATION

1/3 TURN TO AI

THE TRANSLATION GAP

Building the technical bedrock for AI-assisted dermatology



Shifting focus from model accuracy to human decision-making

State 1: Identification

What is it?

Previous studies show internet tools help users identify conditions.

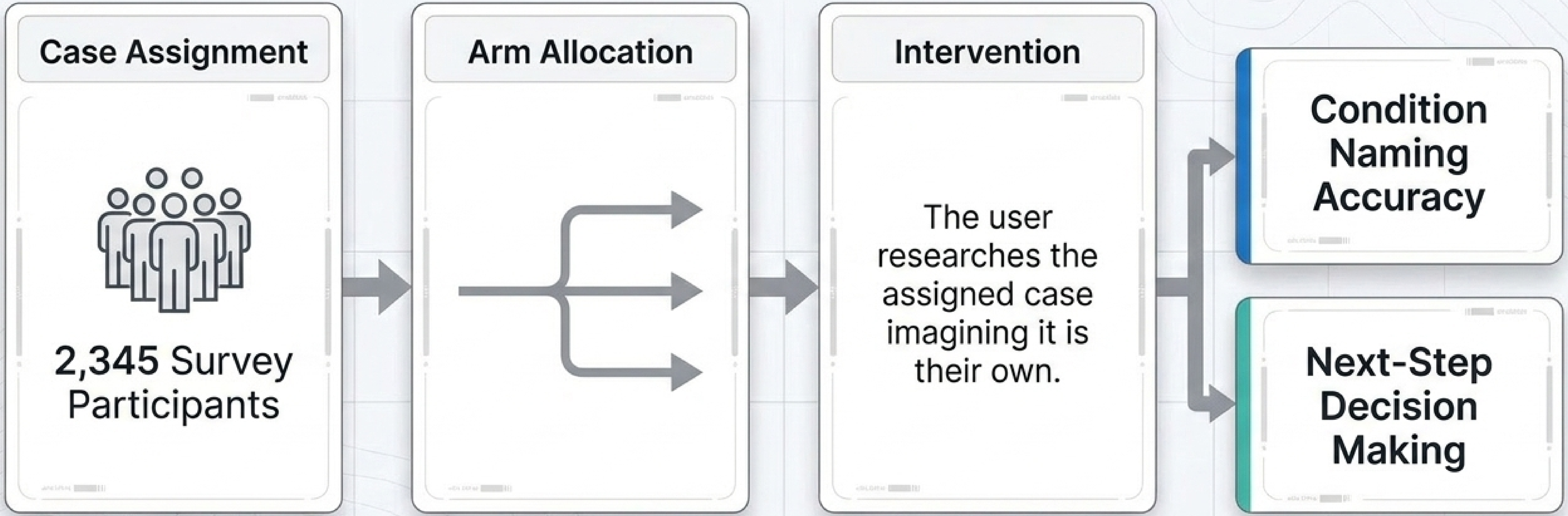
State 2: Triage

What do I do?

Identifying a condition does not necessarily improve the user's ability to decide on medical next steps.

The core research objective is evaluating how structured UI interaction changes a user's ability to both identify a condition and determine their next steps.

A large-scale evaluation of structured AI assistance



Presented with retrospective, de-identified skin cases (images + structured medical history).

Three distinct research arms isolate the impact of AI

Negative Control
Standard Web Search

Experience: Existing tools users are familiar with.

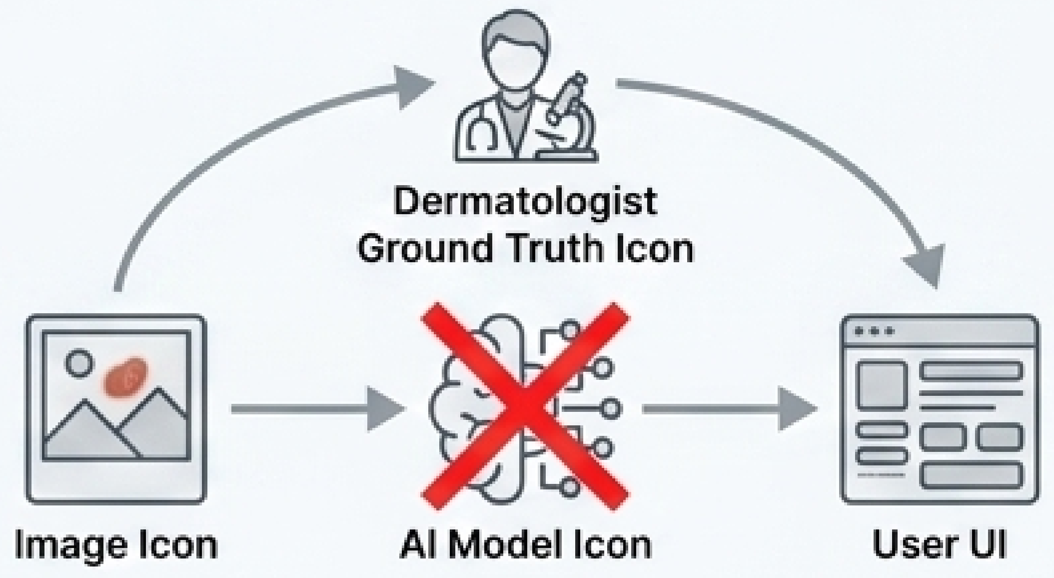
AI Prototype
AI Assistance

Experience: Scrollable carousel of 3-7 matching conditions based on an AI model.

Positive Control
The Wizard of Oz

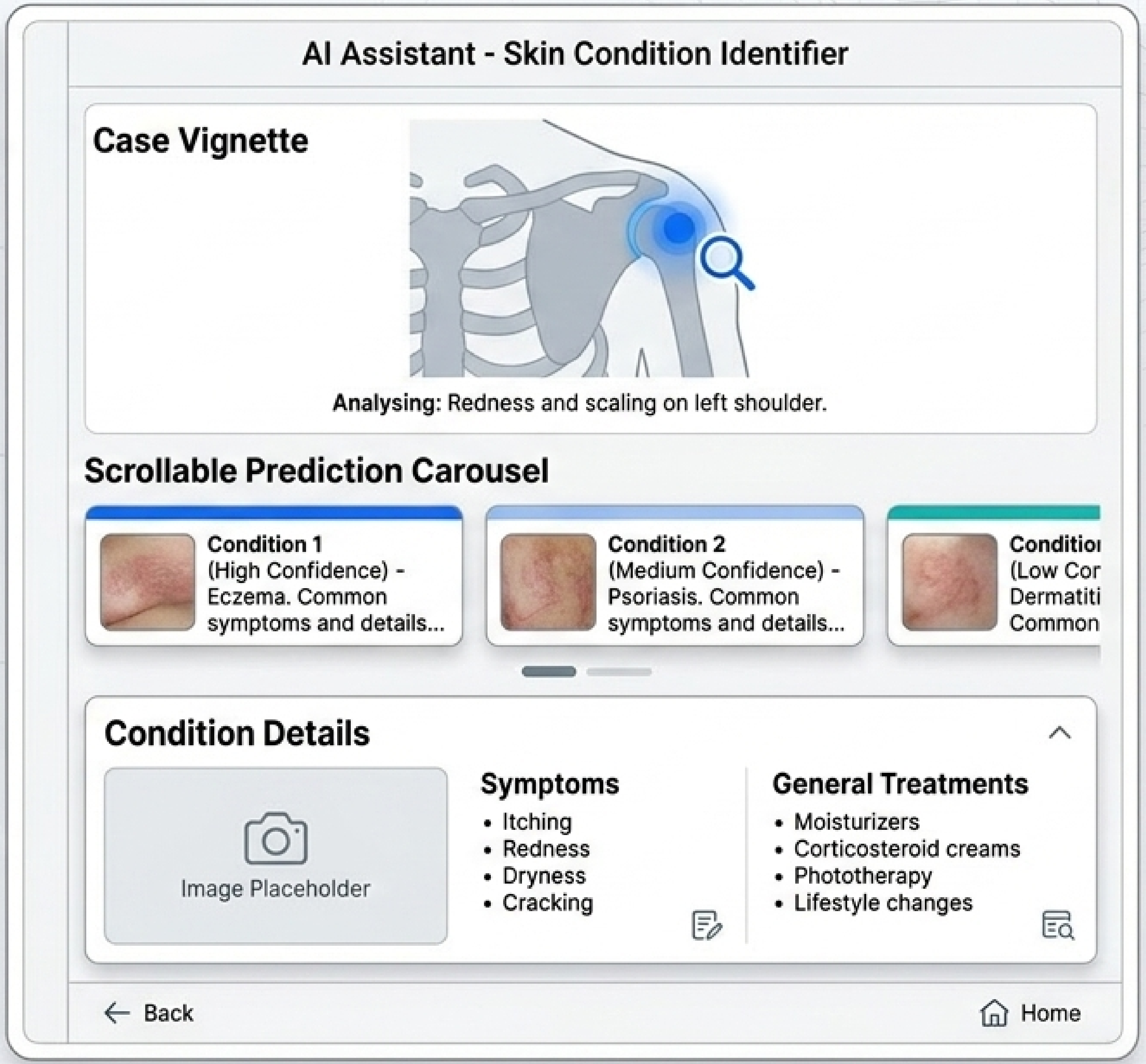
Experience: Mimics perfect AI.

The Wizard of Oz Mechanism



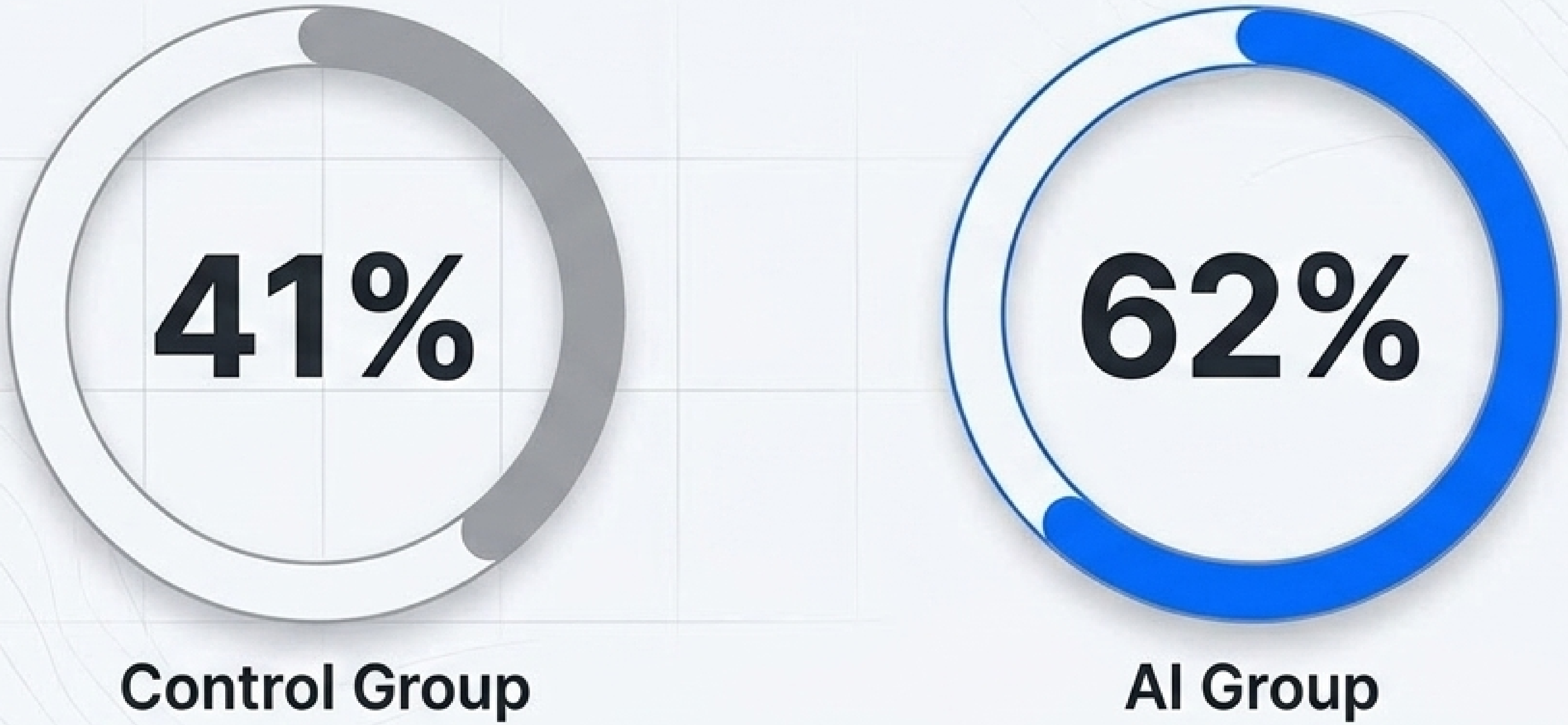
Structuring medical information for the layperson

Designed to be informational, not prescriptive. Focuses on matching images to conditions, leaving the user to interpret next steps.



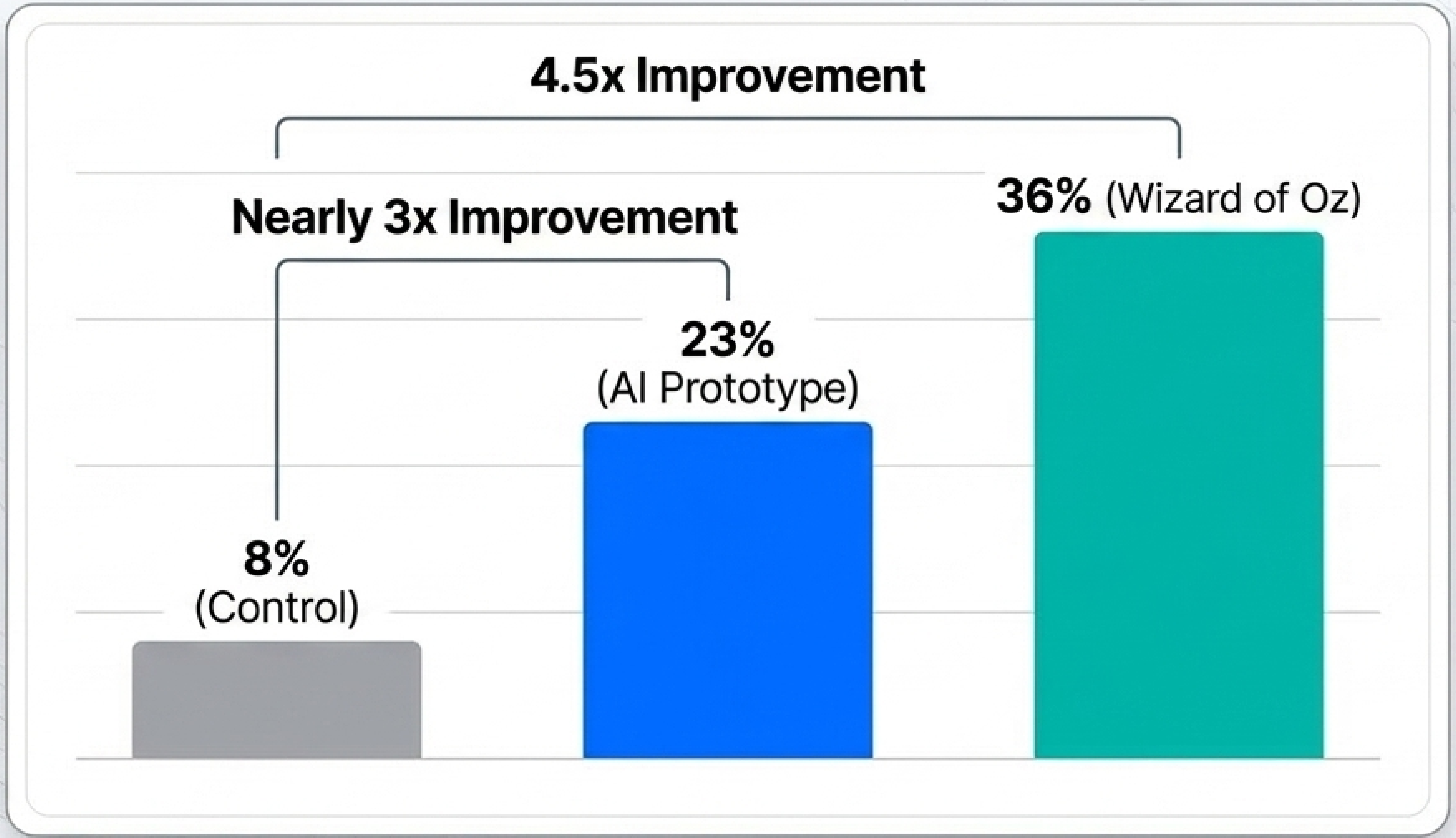
NotebookLM

AI assistance drives a significant leap in user engagement



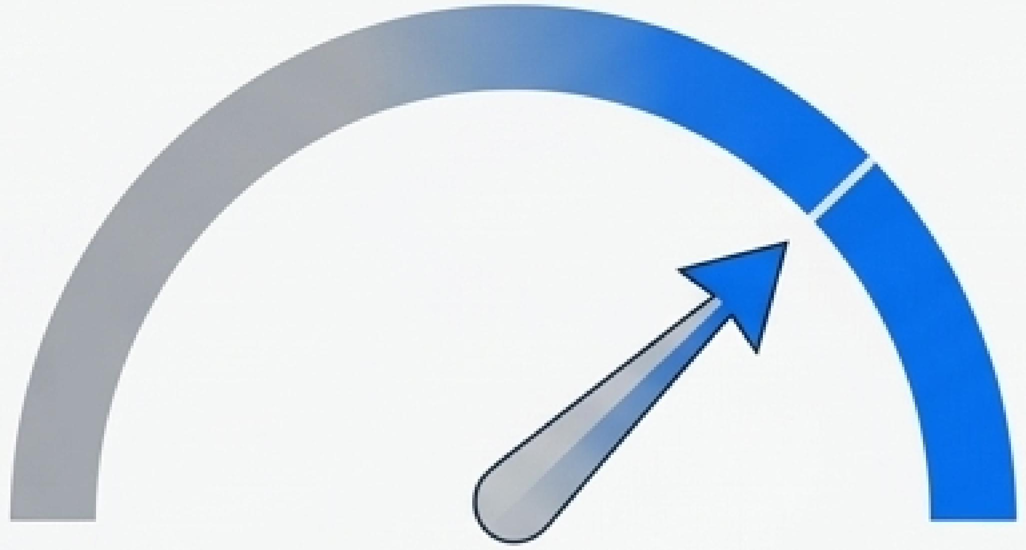
When utilizing the AI tool, users were significantly more **willing to attempt naming the condition shown** compared to standard unassisted search tools. The **interface lowered the psychological barrier to engagement.**

Condition naming accuracy dramatically improves with AI



Note: Accuracy required both the willingness to guess and a match with the dermatologist's differential.

Structured condition cards impart confidence and satisfaction

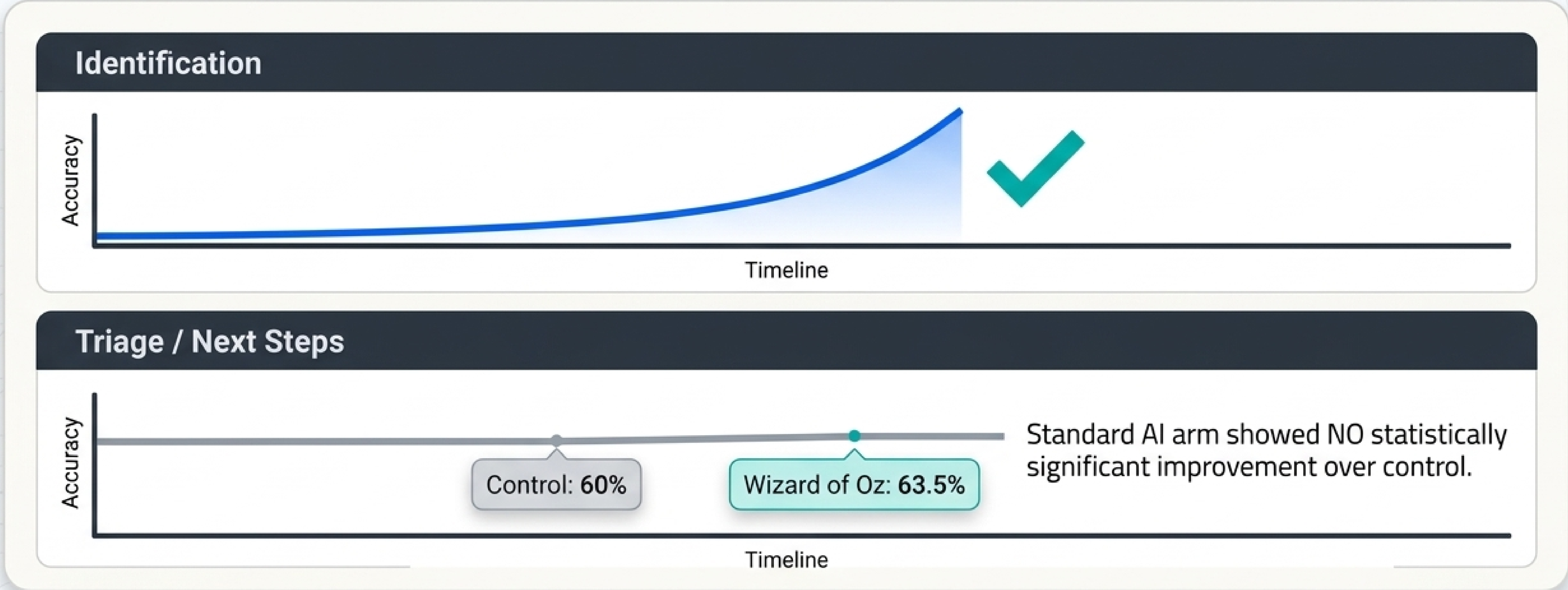


User Confidence

Search Satisfaction

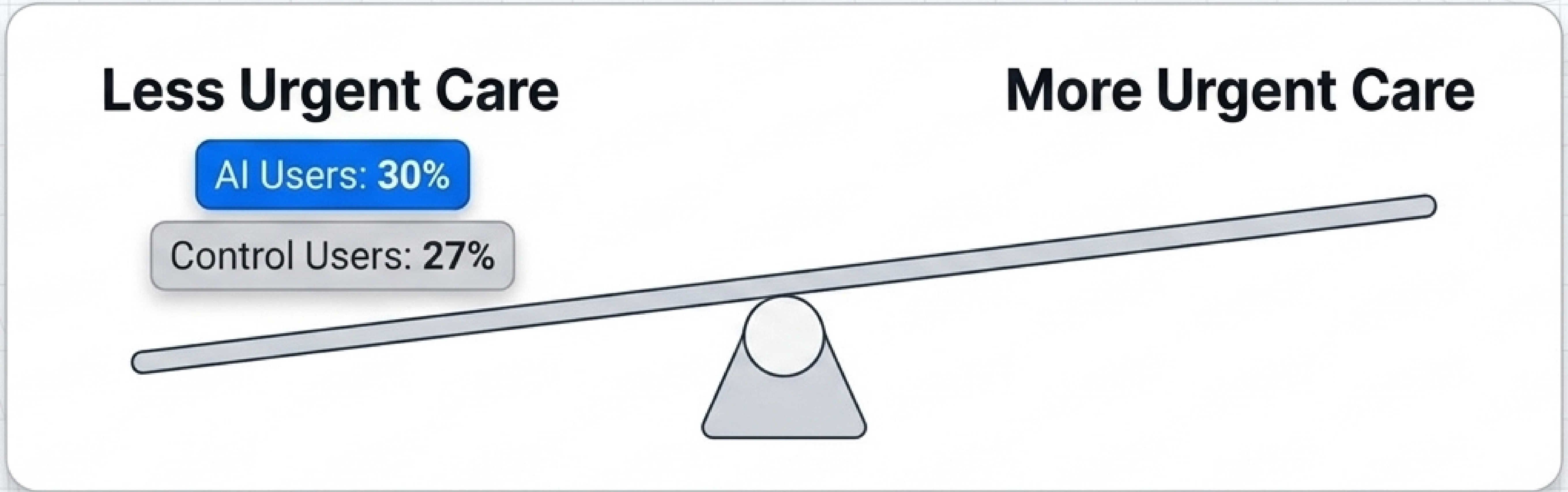
Takeaway: The presentation format **directly impacted** the **psychological safety and satisfaction** of the search experience.

The Identification vs. Triage Gap



Insight: AI highly optimizes the user's ability to answer 'What is it?', but fails to help them answer 'What do I do about it?'

Generic information fails to convey case-specific urgency



The Technical Challenge

Information provided by the AI was authoritative, but written based purely on the condition name. It was completely lacking the patient-specific severity context of the individual case vignette.

Grounding AI development in real-world, diverse communities

The Context

Safety net community (Medi-Cal)

110 consented participants with active skin concerns

App localized into 4 languages

The Outcomes

Naming: 260% increase in ability to name condition.

Clinician Agreement: Clinicians agreed with app predictions **86%** of the time.

Utility: Clinicians found it a helpful shared reference point **92%** of the time.

Key Insight: Participants relied heavily on visual pattern matching. This proves the absolute necessity of diverse skin-tone, severity, and body-part representation in training data.

The Multimodal Solution: Bridging identification and action

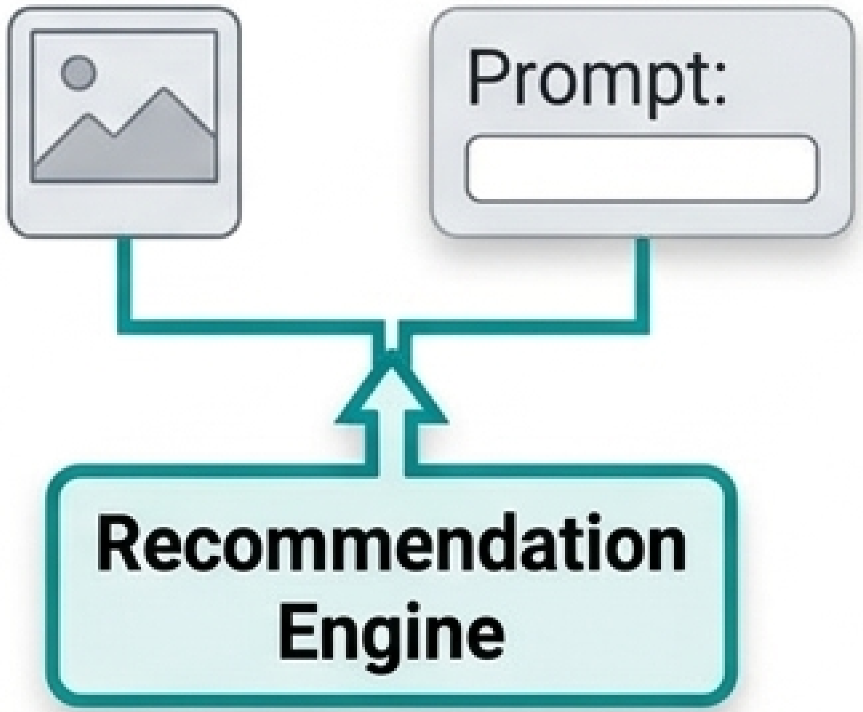
Phase 1: Text Alone
(Standard Search)

Low accuracy, low triage.

Phase 2: Image Alone
(Current AI Prototype)

High accuracy, poor triage (due to generic condition mapping).

Phase 3: Multimodal
(Image + Query-Specific Text)



Takeaway: Combining visual pattern matching with query-specific, severity-aware text is preferred by laypersons and is required to close the triage gap.

Lowering the barrier to entry, guiding the journey contextually

Point 1: The Visual Start.

Image-based AI successfully lowers the barrier to entry, allowing users to accurately name conditions without knowing medical terminology.

Point 2: The Contextual Guide.

Moving forward, personalized, multimodal AI is required to navigate complex medical information safely.

Building highly effective clinical tools requires continuous, human-centered research to ensure that everyone can interpret medical information and safely navigate their healthcare journeys.